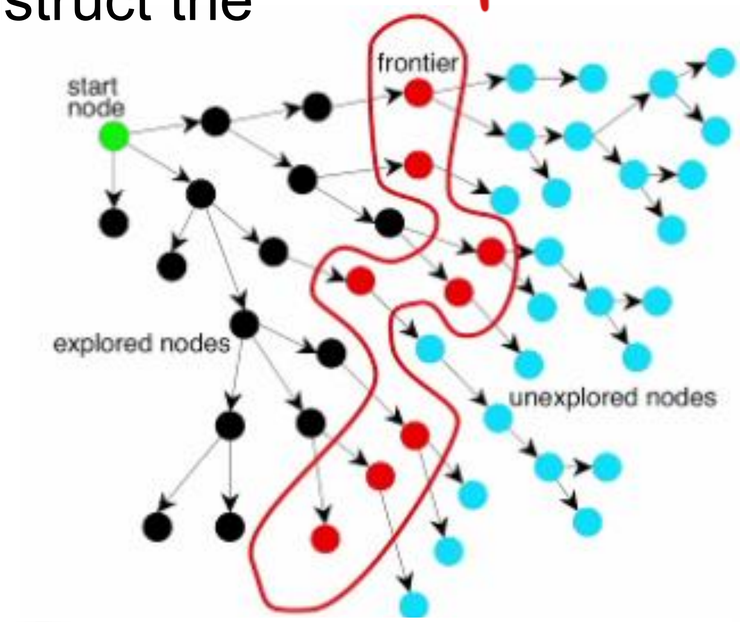
Recap: Frontier---Tracks the boundary between explored an unexplored

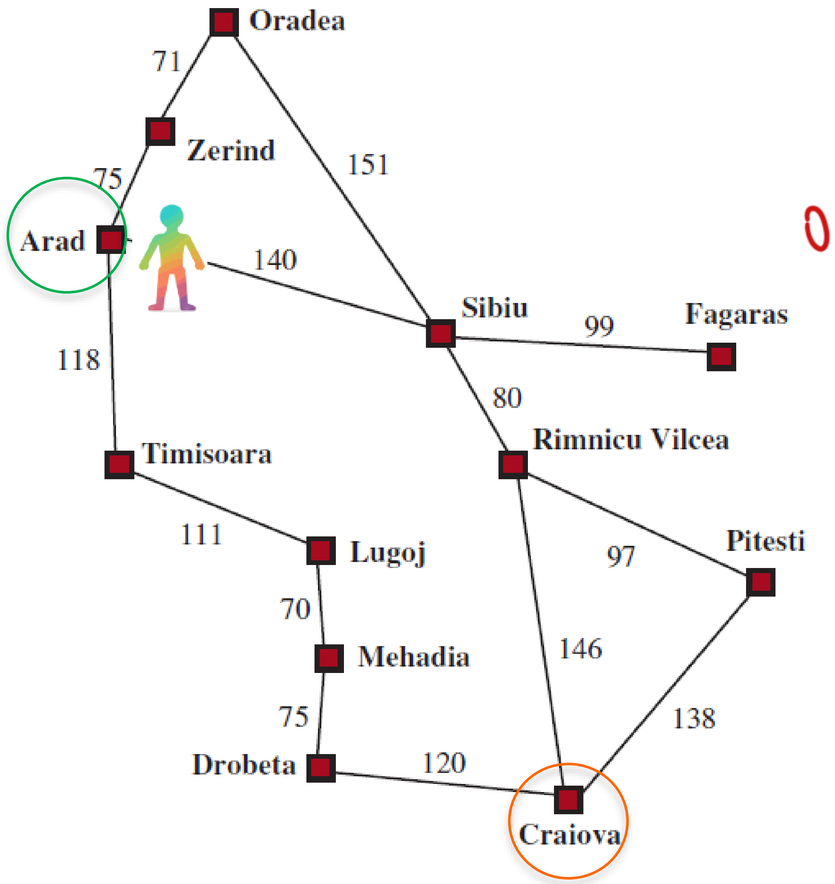
Boundary of explored and unexplored

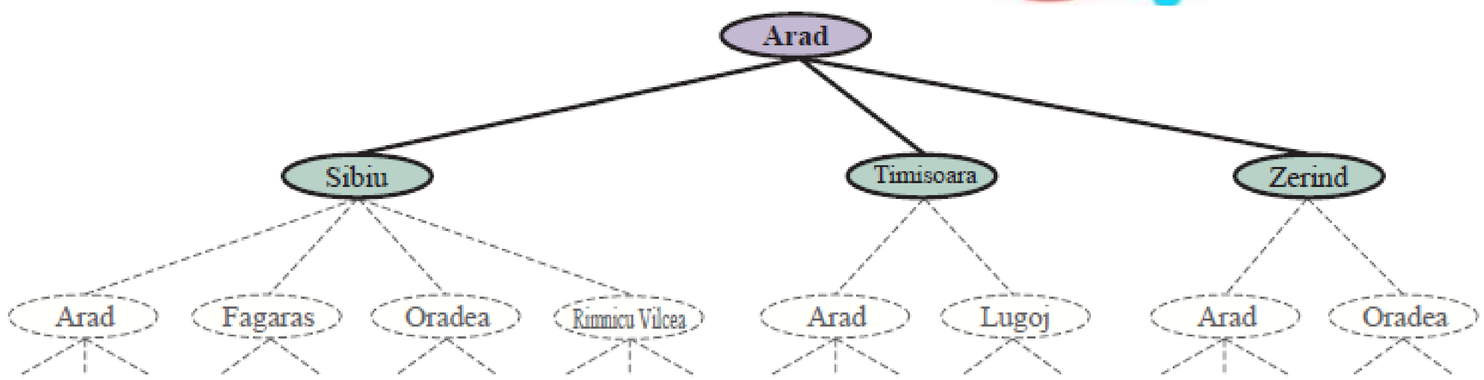
Frontier

- Queue (or other structure) of nodes **waiting to be expanded**
- **Stores state and path information**, needed to reconstruct the solution once the goal is reached



plan to expand 1 of them in next step





Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P90.
Reference: https://www.youtube.com/watch?v=xoqcnWXTXcI

Recap: Graph search vs Tree search

Tree search

```
function TREE-SEARCH(problem, strategy) return a solution, or failure
        Initialize frontier to the initial state of the problem
        loop do
                if the frontier is empty then return failure
                choose a node from the frontier for expansion according to strategy,
                and remove it from frontier
         if the node contains goal state then return the corresponding solution
                else expand the node, and add the resulting child nodes to the frontier
         end
```

Graph search

```
function GRAPH-SEARCH(problem, strategy) return a solution, or failure
        Initialize frontier to the initial state of the problem
        Initialize the explored set to be empty
        loop do
                if the frontier is empty then return failure
                choose a node from the frontier for expansion according to strategy,
                and remove it from frontier
         if the node contains goal state then return the corresponding solution
         add the node to the explored set
         expand the node, and add the resulting nodes to the frontier
            only if not in the frontier or explored set
         end
```

Reference: https://www.youtube.com/watch?v=Y1VywhuRFIs

Summary: BFS vs DFS

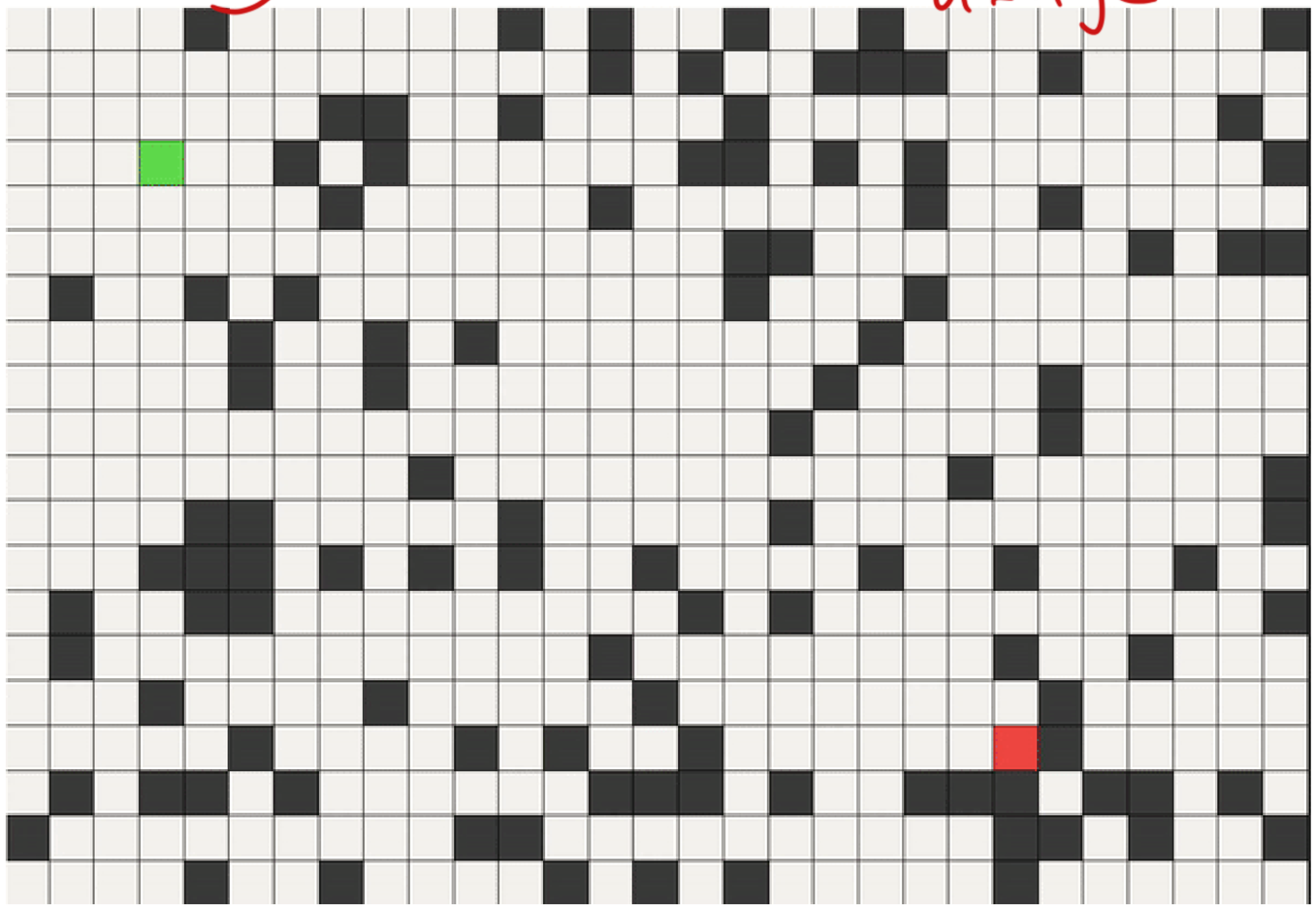
Feature	BFS (Breadth-First Search)	DFS (Depth-First Search)
Search style	Explores all nearby nodes first (level by level)	Goes as deep as possible before backtracking
Data structure used	Queue (FIFO - First in, first out)	Stack (LIFO - Last in, first out)
Will it always find a solution if one exists? (Completeness)	**Yes, if a solution exists**	**Not guaranteed** – may get stuck in deep or infinite paths
Will it find the shortest solution? (Optimality)	**Yes, if all steps cost the same**	**No, may find a longer or less efficient path**
Time needed	Slower if the solution is deep	Can be faster if the solution is deep
Memory needed	High – remembers all paths at one level	Low – tracks only one path at a time
Best for	Finding shortest paths (e.g., in maps) Small search space	Exploring puzzles or scenarios with deep solutions Large or deep search spaces
When to use	Need shortest route/minimum hops Equal step cost Enough memory	Memory is limited When solution may be far/deep Don't need shortest path

More Comprehensive Use less memory

Depth-first search (DFS) in maze

Similar to human play: Early win, small memory usage

- Expand the deepest unexplored node in the frontier first.
- *Explore one path as far as it can go before backtracking.*
- All the successors of a node are expanded before moving to other siblings.

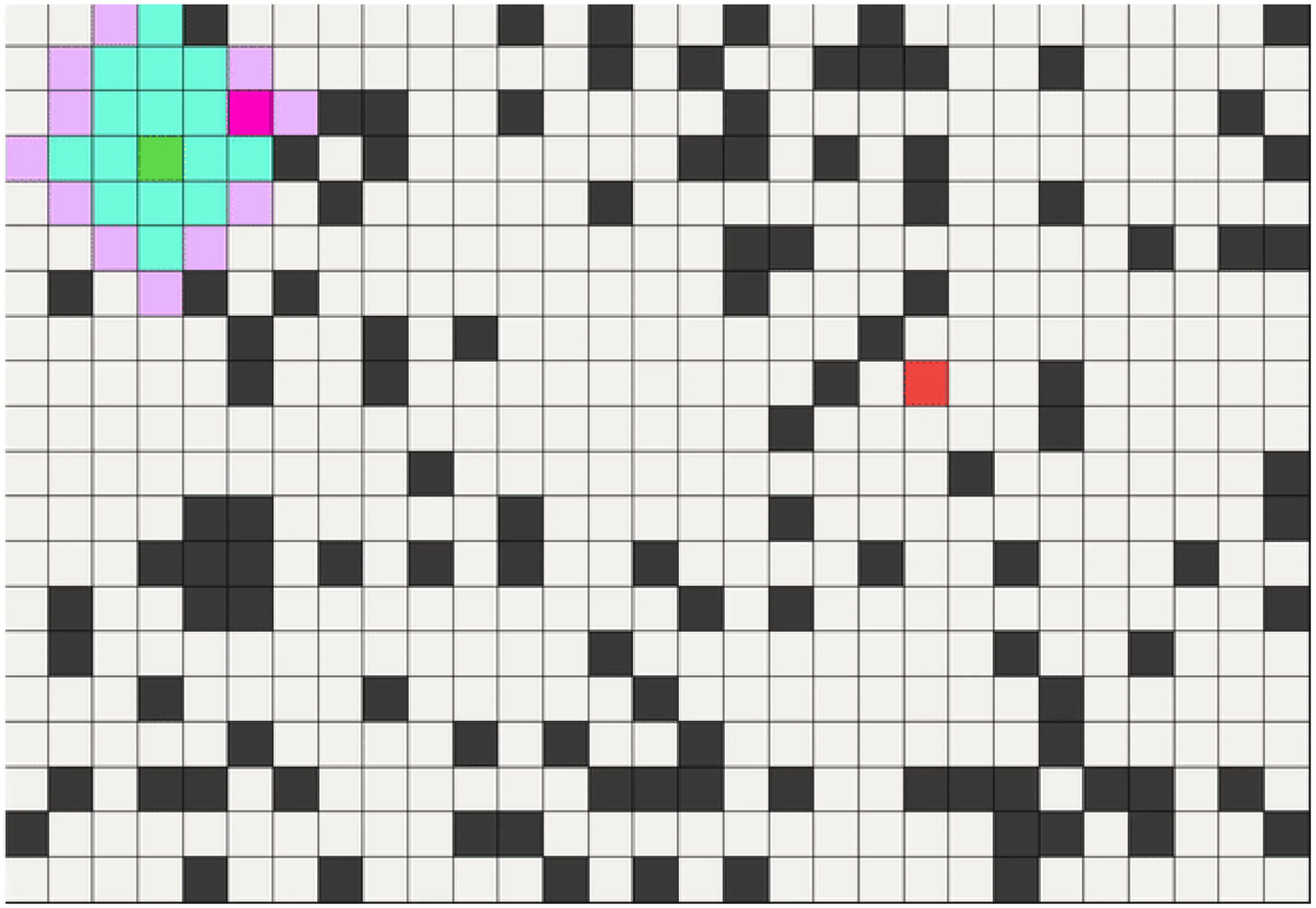


Not neccessary the shortest path

Source: https://medium.com/codex/python-project-idea-graph-traversal-and-pathfinding-algorithm-visualisations-99595c414293

Breadth-first search (BFS) in maze

- Explore nodes in “layers” (level by level)
- Can find everything findable from a given starting node
- *Can compute minimum hops between two nodes (i.e., shortest path, if all steps cost the same)*



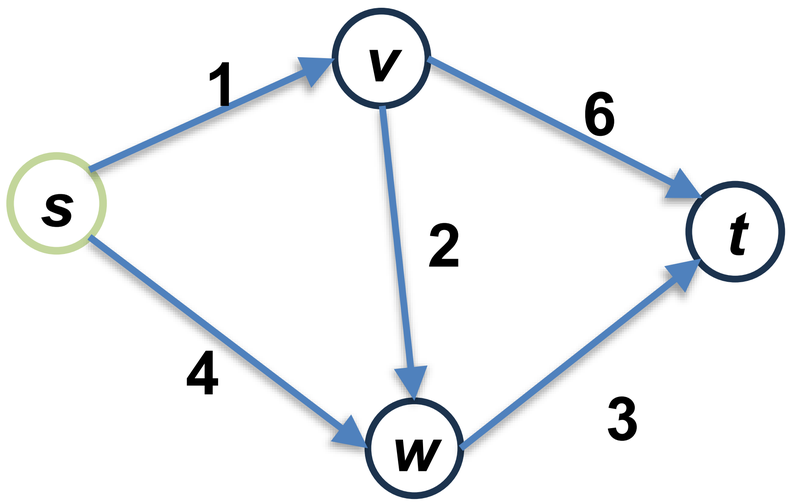
What if step costs are different?

Source: https://medium.com/codex/python-project-idea-graph-traversal-and-pathfinding-algorithm-visualisations-99595c414293

Quick question 1

Can we use BFS to find the shortest path from node *s* to node *t* ? The numbers on each edge represent the path distance of that edge.

We can't use BFS to solve the problem that the cost for each step isn't equal.



Limitation of BFS

BFS is effective for finding the shortest paths only in graphs where all the edges have the same (unit) length.

***Uniform-Cost Search* (UCS, also known as Dijkstra's algorithm)** extends this by handling graphs with any *non-negative* edge length.

**Uniform-cost search (UCS) is the name used in the AI community, while the theoretical computer science community refers to it as Dijkstra's algorithm.*

Agenda

Uniform ≠ Uninformed

- Uniform-cost search (Dijkstra's algorithm)
- Informed search (greedy best-first, A*)

**The uniformity is in the way the algorithm fairly (uniformly) evaluates and expands paths based on cost, rather than by depth or number of steps.*

Dijkstra's algorithm & BFS

- Dijkstra's algorithm extends BFS for varying edge lengths

- Have a shared principle

- Similarity with equal edge lengths

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P94 – P96

Dijkstra's algorithm: Overview

Notation:

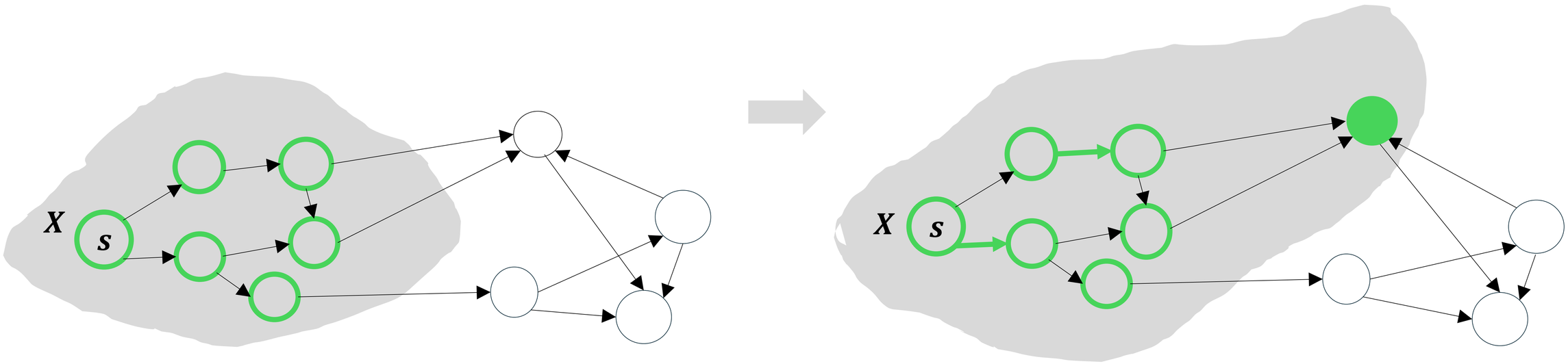
$X = \{\}$ [explored set: nodes processed so far]

$Q = \{\}$ [frontier: nodes waiting to be expanded]

$dist[v]$ [shortest path distances from the start s to v]

$e(v, w)$ [edge from v to w]

$l(v, w)$ [length/cost of $e(v, w)$]



Dijkstra's algorithm: Overview

Initialize:

$Q = \{s\}$ [start from initial sate s]

$dist[s] = 0$ [computed shortest path distances]

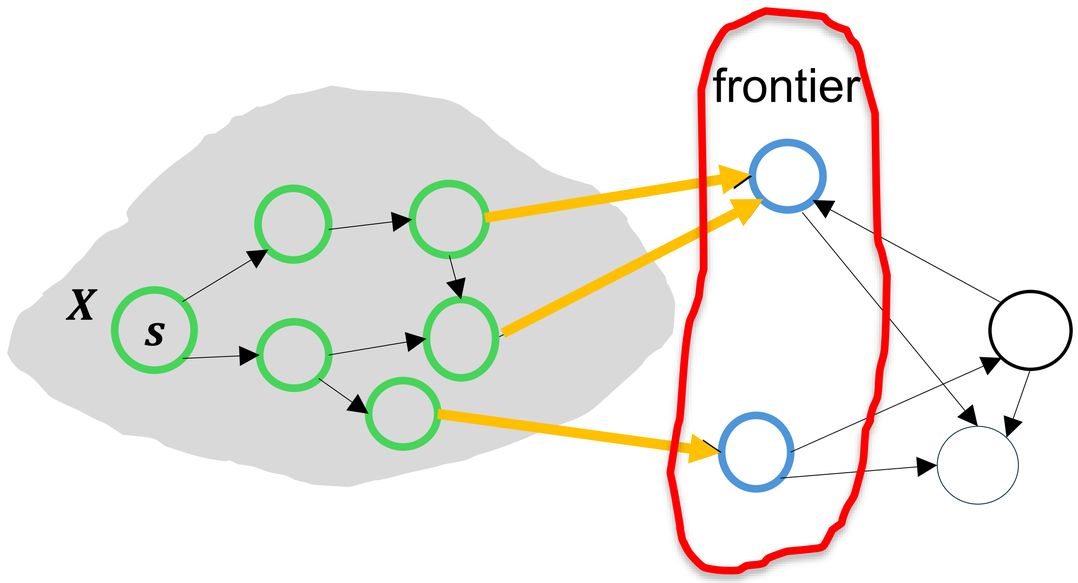
Dijkstra's algorithm: Overview

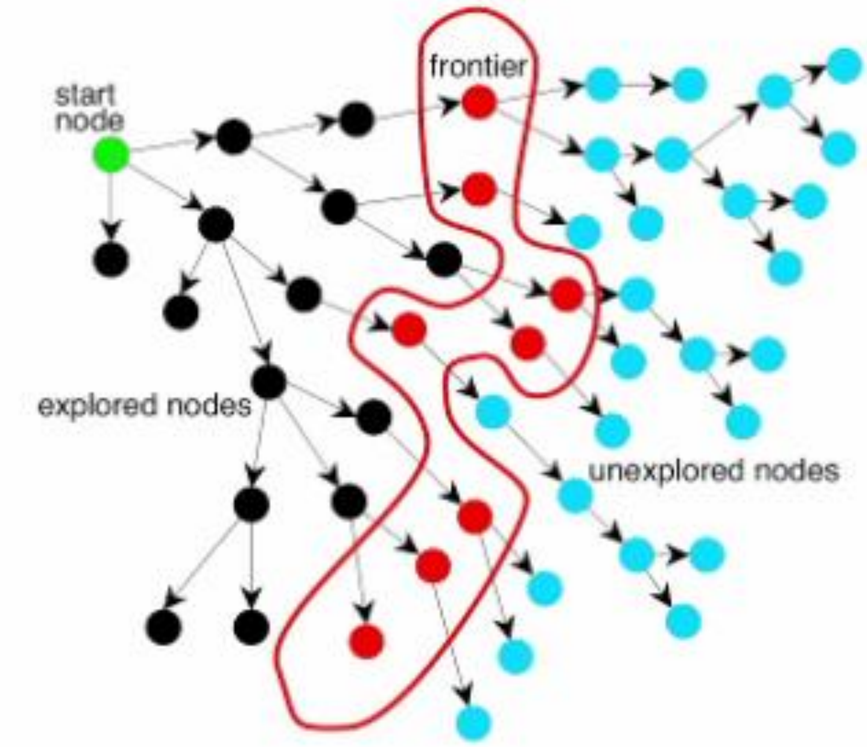
Loop:

Among all $e(v,w) \in E$, with $v \in X, w \notin X$,

pick the one that minimizes $dist[v] + l(v,w)$ *(Dijkstra's greedy criterion)*

From the frontier, choose the node closest to the source.





Dijkstra's algorithm: Overview

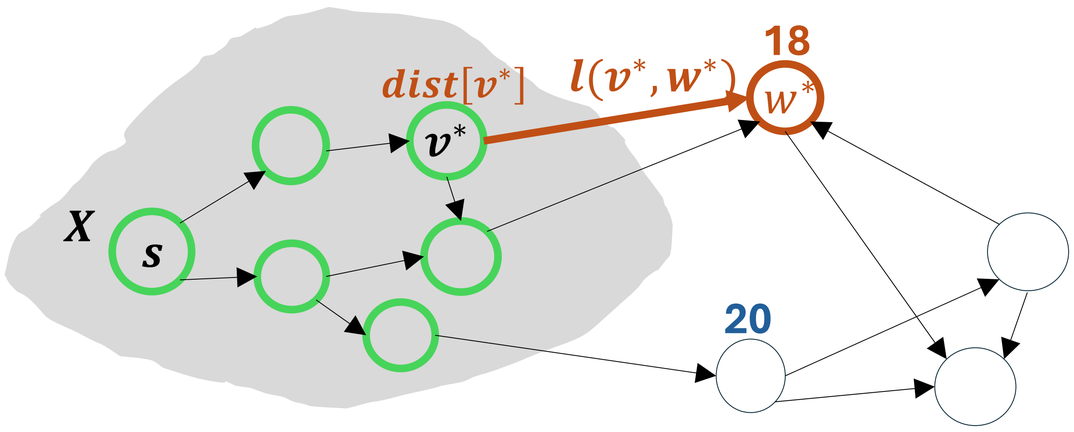
Loop:

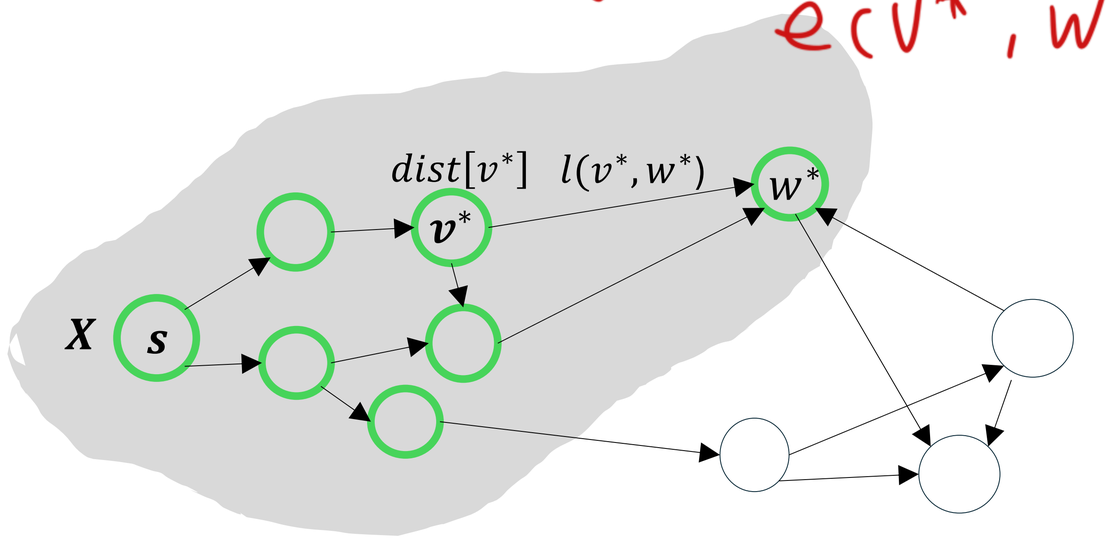
Among all $e(v,w) \in E$, with $v \in X, w \notin X$,

pick the one that minimizes $dist[v] + l(v,w)$ *(Dijkstra's greedy criterion)*, add w^* to X, $dist[w^*] = dist[v^*] + l(v^*,w^*)$

known

length (cost) of $e(v^*, w^*)$

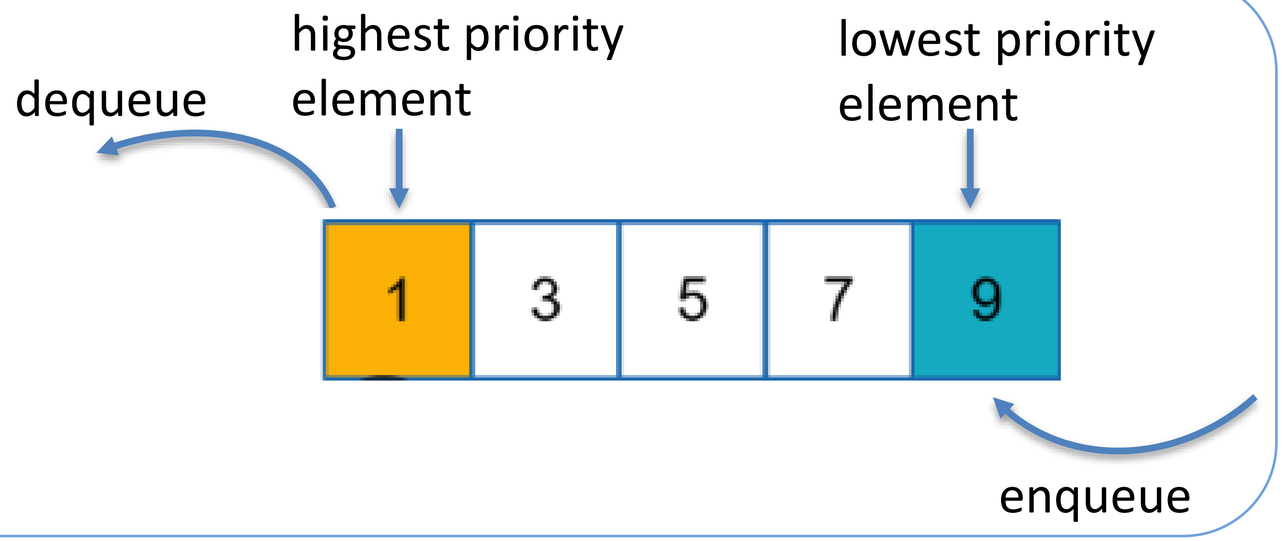




Dijkstra's algorithm: Frontier as a priority queue

- Dijkstra's algorithm always expands the frontier node with the *smallest distance from the source (initial state)*.
- In practice, the **frontier** is managed as a **priority queue**, so nodes in the frontier are expanded in order of their current *shortest* distance from the source.

Priority queue:



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P94 – P95.

Dijkstra's algorithm (uniform-cost search): Pseudocode (v2)

- Start with a **frontier (priority queue)** containing the initial state (source), with path cost = 0.
- Start with an **empty explored set**.
- Repeat:
  - -- If the frontier is empty, then no solution.
  - -- Remove the frontier node with the lowest path cost from the source.
  - -- If this node is the goal, return the solution.
  - -- Add the node to the explored set.
  - -- Expand the node:
    - For each child, compute its total path cost from the source.
    - If the child is not in the frontier or explored set, **add it to the frontier** with its cost.
    - If the child is already in the frontier with a **higher cost**, update it with the **lower cost**.

closest node to the source

If the child in the explored set, we don't process it.

If the new path is better than the old path, then discard the old one.

BFS vs DFS vs Dijkstra's algorithm

Feature	BFS (Breadth-First Search)	DFS (Depth-First Search)	Dijkstra's (Uniform-cost search)
Search style	Explores all nearby nodes first (level by level)	Goes as deep as possible before backtracking	Expands the node with the *lowest total cost* from the source (initial state)
Data structure used	Queue (FIFO - First in, first out)	Stack (LIFO - Last in, first out)	Priority queue (based on path cost)
Will it always find a solution if one exists? (Completeness)	Yes, if a solution exists	Not guaranteed – may get stuck in deep or infinite paths	Yes
Will it find the shortest solution? (Optimality)	Yes, if all steps cost the same	No, may find a longer or less efficient path	Yes, even when edge costs differ
Can it handle different edge costs?	No – assumes all edges have equal cost	No – ignores cost	Yes – works with any non-negative edge cost
Time efficiency	Slower if the solution is deep	Can be faster if the solution is deep	Slower in large graphs with varying costs
Memory usage	High – remembers all paths at one level	Low – tracks only one path at a time	High – must track cumulative costs for all paths
Best for	Finding shortest paths (e.g., in maps) Small search space	Exploring puzzles or scenarios with deep solutions Large or deep search spaces	Finding the cheapest/least-cost path in weighted graphs (e.g., shortest time, lowest expense)
When to use	Need shortest route Equal step cost Enough memory	Memory is limited When solution may be far/deep Don't need shortest path	Need shortest route with different edge costs; can afford higher memory usage

Uninformed search & informed search

Uninformed search (BFS, DFS, Dijkstra's)

- Has no additional knowledge beyond what is given in the problem definition.
- Generate successor states and check whether each is a goal.
- As a result, the search proceeds blindly—systematically exploring the space until the goal is found.
- Search methods are distinguished by the order in which the nodes are expanded.

Informed search (greedy best-first, A*)

- Uses additional knowledge (heuristic) to estimate how close a state is to the goal.
- The heuristic guides the search, often reducing the number of nodes explored and improving efficiency.

Greedy best-first search (GBFS)

An informed search algorithm that expands the node that *appears to be closest to the goal*, as estimated by a heuristic function $h(n)$

$h(n)$: Estimated cost of the cheapest path from the current node n to a goal. (for goal node: $h(n) = 0$)

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P102 – P103

Heuristic function

Heuristic knowledge is useful, but NOT always accurate.

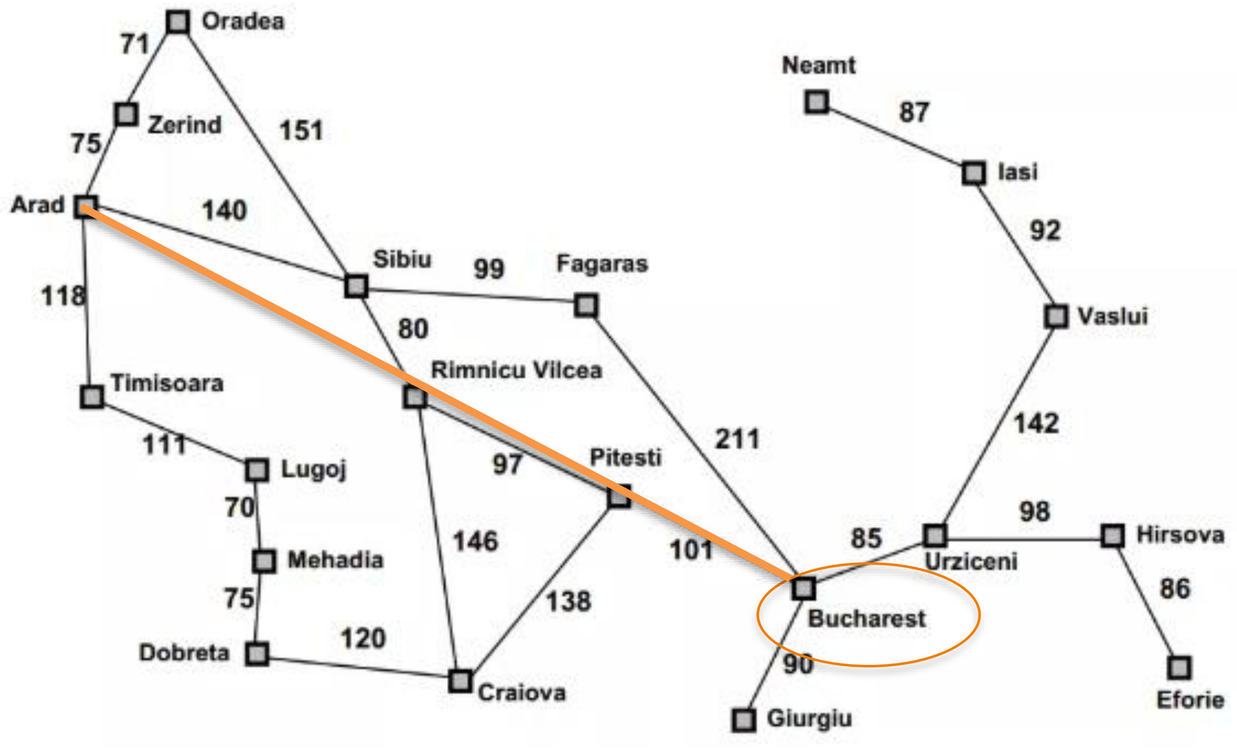
Heuristic algorithms use heuristic knowledge to guide the search process.

A **well-designed heuristic** can significantly reduce the number of nodes explored, making the search more efficient.

Example: Heuristic in path planning

In map navigation, a common heuristic is the straight-line (Euclidean) distance to the goal.

This provides a reasonable estimate of how far a place is from the destination.



Straight–line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

$h(n)$

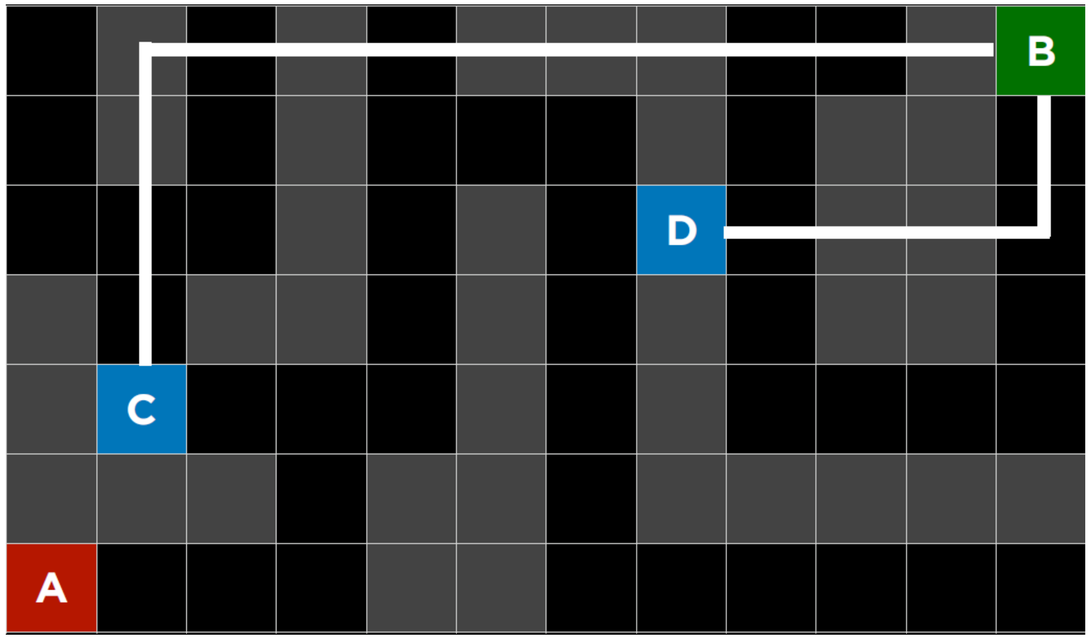
For two points $p = (x_1, y_1)$ and $q = (x_2, y_2)$, the Euclidean distance (also known as L2 distance) is:

$$d_{Euclidean}(p, q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

The heuristic is NOT always accurate, as actual travel paths are rarely straight due to roads, obstacles, and terrain.

Example: Heuristic in grid maze

Heuristic used: Manhattan distance (also known as L1 distance), which sums horizontal and vertical steps to the goal.

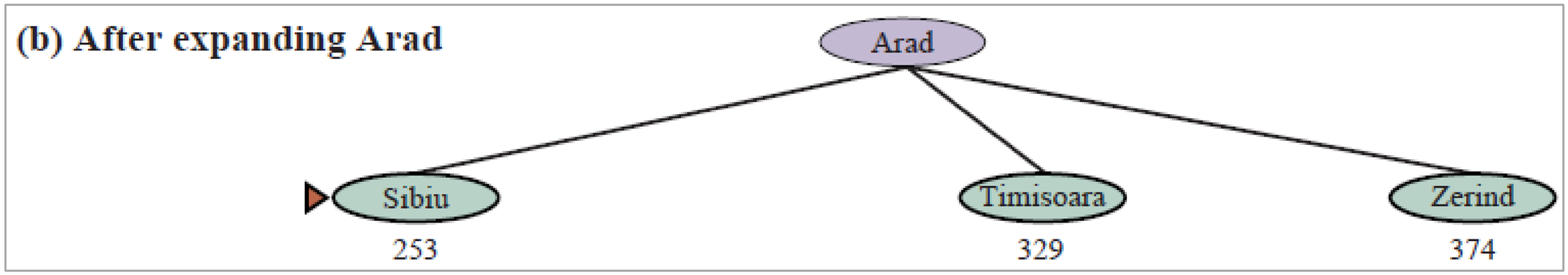


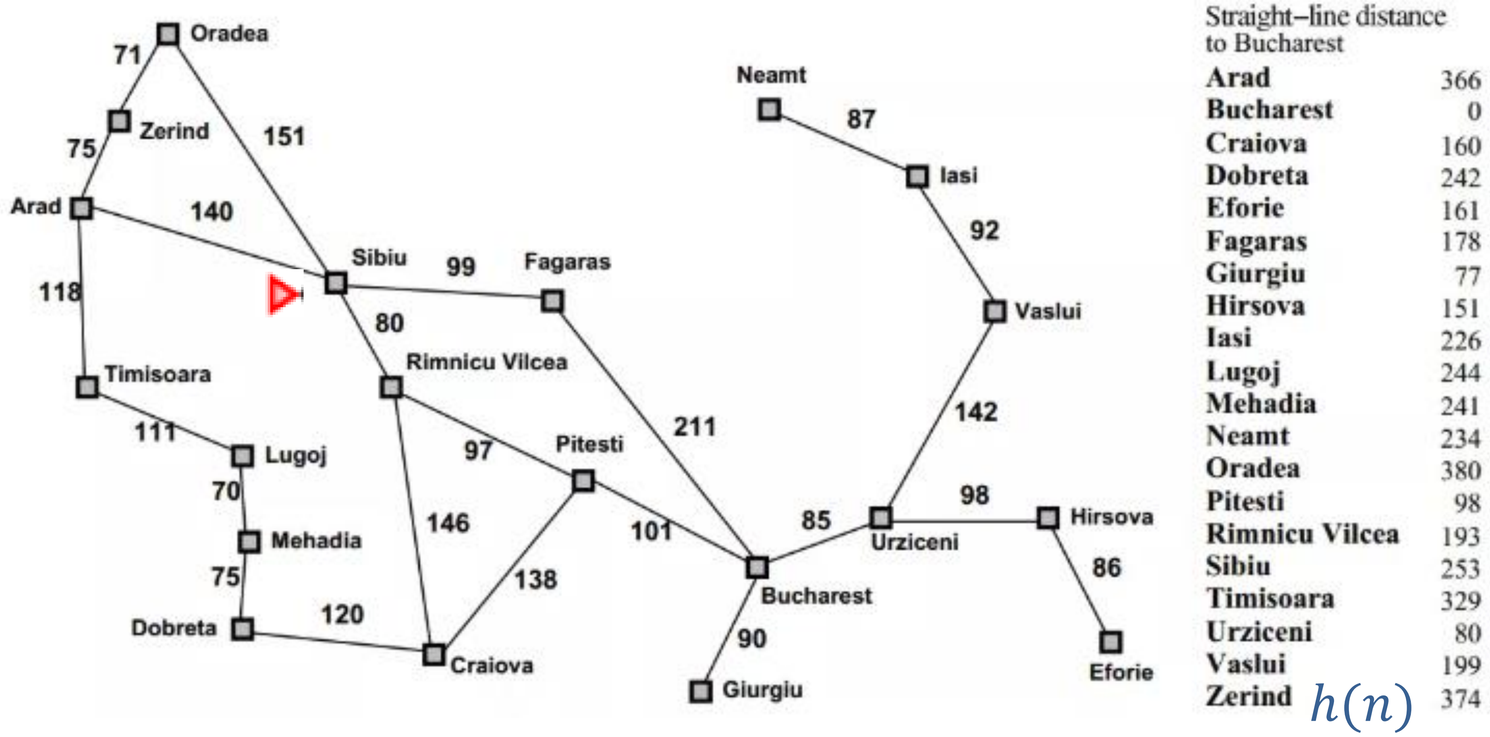
For two points $p = (x_1, y_1)$ and $q = (x_2, y_2)$, the Manhattan distance is:

$$d_{Manhattan}(p, q) = |x_1 - x_2| + |y_1 - y_2|$$

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

Path planning with greedy best-first



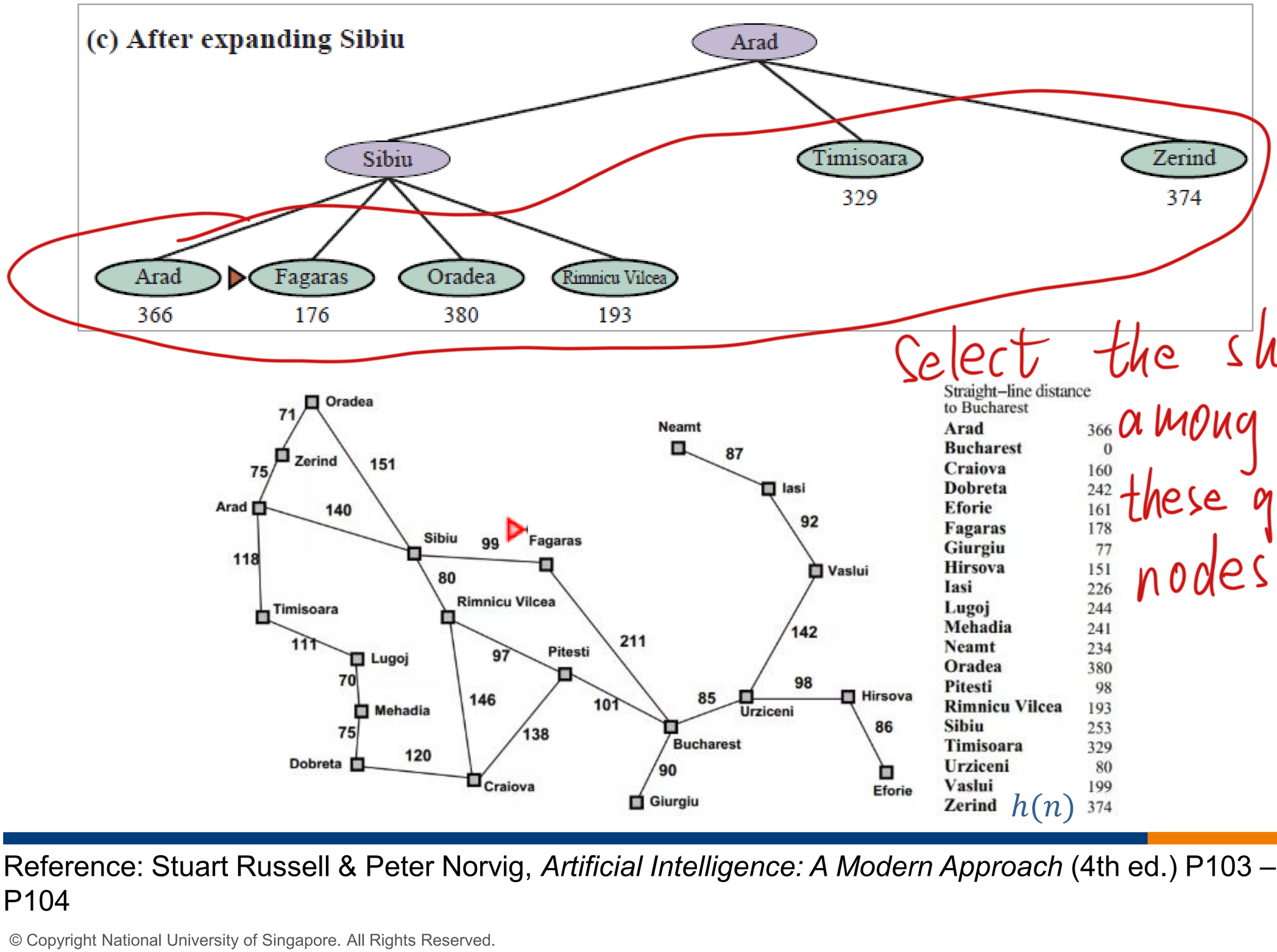


Straight–line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

$h(n)$

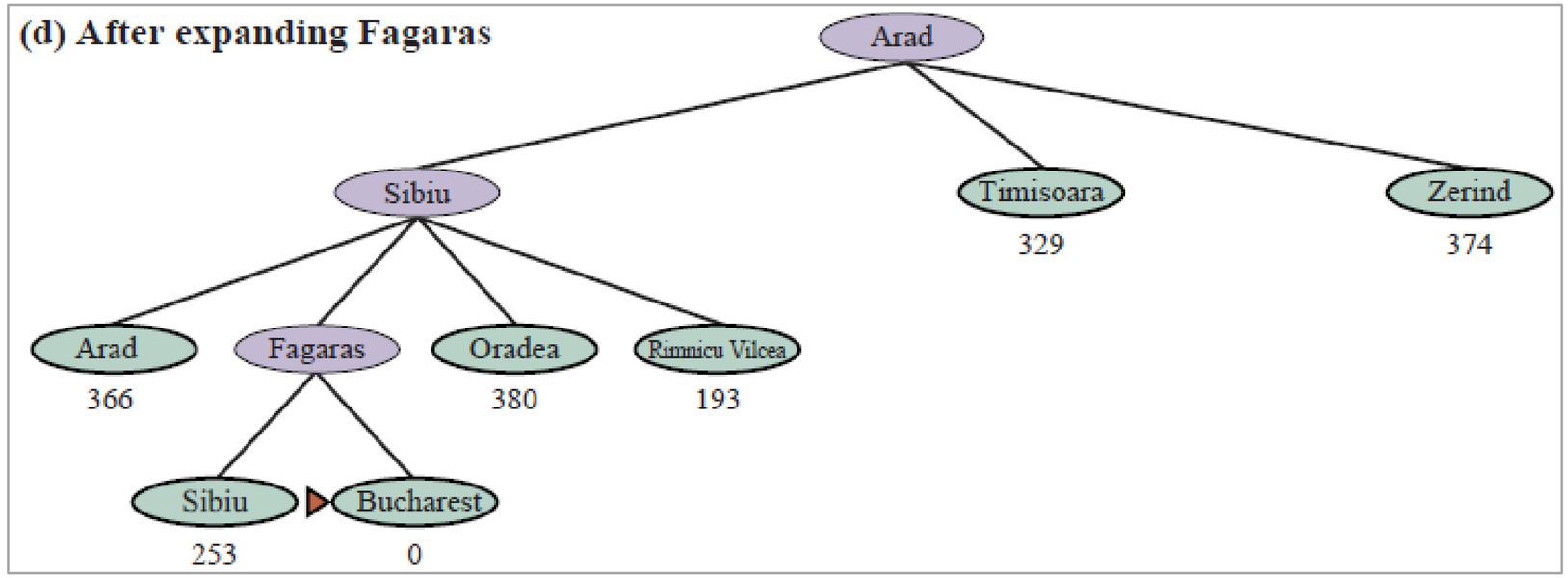
Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P103 – P104

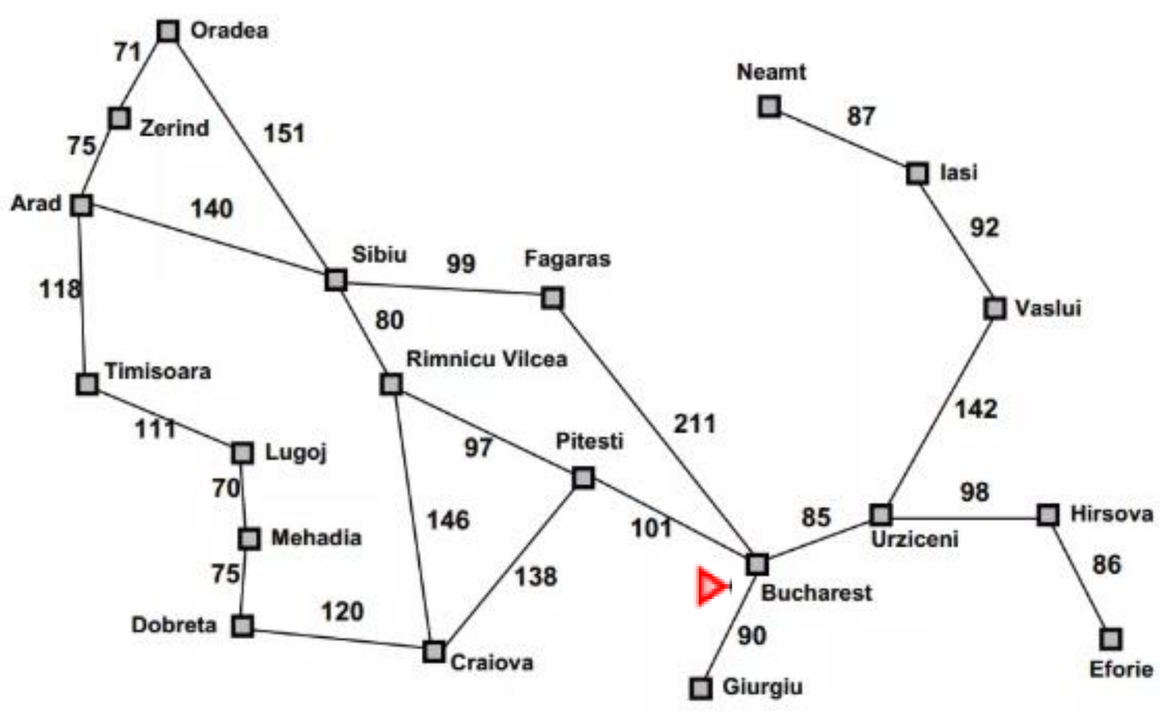
Path planning with greedy best-first



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P103 – P104

Path planning with greedy best-first





Straight-line distance to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

$h(n)$

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P103 – P104

A* search: A Preview

- A* search addresses the limitation of greedy best-first search by also considering the cost so far, not just the heuristic.
- Perhaps the most well-known search algorithm in AI and pathfinding

- Widely used in navigation systems like Google Maps, Apple Maps, and other routing applications

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P103 – P109